PROJECT SYN

Real-Time Disaster Healthcare Command Center for Mega-City Crises

Sub-Millisecond Triage, Dynamic Flood Corridors & Mainframe Autonomy | IBM Z Datathon 2026

Track: Real-Time AI for Critical Decisions | Team: Yuvraj Lokhande, Palak, Anshika, Nidhi, Ritesh, Bhavya

// URBAN PROBLEM STATEMENT // IBM Z DATATHON 2026

The Crisis: Extreme Monsoon Inundation & Systemic Fragility

Mumbai Severe Monsoon Downpour & Coastal Sea Surge Emergency

1. Electrical Grid Vulnerability

GRID OUTAGE: SUBSTATIONS TRIP

- Severe monsoon downpours flood low-lying Dadar & Lalbaug basins.
- BEST 22kV substations trip, forcing trauma centers onto diesel backup.
- ICU ventilators and oxygen generators face acute runtime depletion.

2. Arterial Corridor Submersion

CORRIDOR BLOCK: CRITICAL

- Hindmata & Milan subways submerge under standing floodwaters.
- Ambulances carrying critical trauma patients trapped in dead-end gridlocks.
- Consumer GPS apps route vehicles into drowned underpasses with 0 visibility.

3. Adversarial Cyber Poisoning

DATA INTEGRITY: VULNERABLE

- Unauthenticated IoT telemetry risks corrupting hospital bed registries.
- Spoofed packets threaten to divert emergency convoys to closed facilities.
- Public cloud telecom lines flood, cutting off centralized cloud AI.

// SOLUTION ARCHITECTURE // IBM Z DATATHON 2026

Dual-Mainframe Safety-Bounded Command Mesh

Physical Separation of Edge Verification, AI Inference & Audit Ledger

Pillar 1: Hardware Cryptographic Gate

CRYPTO LATENCY: < 0.24 ms

- IBM z16 CPACF wire-speed HMAC-SHA256 verification (<0.24ms).
- Monotonic anti-replay ring buffer drops spoofed packets immediately.
- Zero-Trust Policy: Untrusted telemetry rejected before AI ever sees it.
- Simulated Attack Button in UI proves real-time 401 Unauthorized rejection.

Pillar 2: Safety-Bounded Composite AI

INFERENCE SPEED: < 0.8 ms

- IBM Snap ML Random Forest scores ICU saturation in <0.8ms.
- Deterministic Dijkstra Pathfinder recalculates elevated flyover bypasses.
- No hallucinated routes: Probabilistic LLMs strictly forbidden from vehicle dispatch.
- Trilingual local SLM (IBM Granite 1.5B s390x) outputs in Marathi, Hindi, English.

// ALGORITHMIC PRECISION // IBM Z DATATHON 2026

Dijkstra-Pathfinder: Saving Ambulances from Drowning

Dynamic Flood Penalties & Elevated Flyover Routing across 35 Mumbai Hospitals

35 Mumbai Trauma Centers

35 HOSPITALS MAPPED

- Enriched with OpenCity CKAN municipal health data (Resource 8d1c157e).
- Monitors KEM, Tata Memorial, Sion LTMGH, Nair, Lilavati & Bombay Hospital.
- Real-time ICU ward capacity and backup generator telemetry.

Elevated Flyover Bypass

TRANSIT GAIN: -42 MIN

- Automated weight penalty applied to flooded surface streets.
- Reroutes transit dynamically onto Dr. Babasaheb Ambedkar Elevated Flyover.
- Cuts emergency patient transit time by 42 minutes during peak storm surge.

Trilingual Grassroots Alerts

EN / HI / MR DISPATCH

- Dispatches actionable tactical briefs natively in Marathi, Hindi & English.
- Telegram Webhook Bot API with exponential backoff.
- Simulated LoRa SX1276 radio pings for air-gapped field paramedics.

PROJECT SYN // DUAL IBM LINUXONE III (s390x) MAINFRAME MESH // LIVE DEMO: 148.100.79.88:8000/ui/

Bare-Metal Mainframe Deployment & Live Validation

Dual IBM LinuxONE III (s390x) Community Cloud Instances

Live Production Mainframes

2 MAINFRAME NODES LIVE

- Node 1 (KEM Hospital Apex): http://148.100.79.88:8000/ui/
- Node 2 (Tata Memorial Hub): http://148.100.98.9:8000/ui/
- Live System Guide & Setup: https://yuvrajlokhande19.github.io/Project-Syn-IBM-LinuxONE/
- Interactive Swagger OpenAPI: http://148.100.79.88:8000/docs
- Dual-node P2P hash pinning ensures continuous zero-downtime failover.

JupyterLab Scientific Suite (Port 38888)

5 NOTEBOOKS VALIDATED

- Running in s390x certified container on Node 1 (Token: Your_Token1).
- Notebook 00: s390x Hardware, Big-Endian & CPACF Verification.
- Notebook 01: OpenCity CKAN Telemetry Ingestion Pipeline.
- Notebook 02: Snap ML Random Forest Anomaly Benchmarks (<0.8ms).
- Notebook 03: Dijkstra Dynamic Green Corridor Rerouting Graph.
- Notebook 04: CPACF 10,000-Packet Wire-Speed HMAC Validation.